



**Department of Software Engineering**

**A PROJECT REPORT ON**

**“THE PROPOSED ESIGELEC HOSPITAL NETWORK  
DESIGN”**

*Submitted by*

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## **ABSTRACT**

This project involves designing a secure, high-performance network infrastructure tailored for a hospital environment. Leveraging Cisco Packet Tracer for simulation, the project addresses the critical demands of modern healthcare, including data security, high availability, and real-time connectivity. The proposed network adopts a three-tier architecture that balances scalability and efficiency across client, application, and database layers, ensuring seamless operation for patient care applications and medical devices. Roles in the IT department are strategically allocated: network administration for reliability, network security for patient data protection and compliance, network architecture for scalability, and IT support for operational continuity. Key project goals include HIPAA-compliant security measures, redundancy to minimize downtime, and robust integration of wired and wireless technologies. The project also outlines future enhancements, such as telemedicine support, data-driven patient care, AI-driven cybersecurity, and cloud integration.

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# CHAPTER 1

## 1 Introduction

In ESIGELEC hospital, an IT department was newly created to internally create and manage the hospital networking needs. The aim of this new development was to reduce the cost of outsourcing the hospital networking needs and also for more control, future scaling and flexibility of the hospital network. Our team is composed of four specialized units, each with a dedicated role to ensure the efficient, secure, and uninterrupted operation of our hospital's network infrastructure. Together, we work to provide reliable network services, safeguard sensitive patient information, and support the day-to-day technical needs of our healthcare professionals.

### 1.1 Various Roles in The Hospital IT Department

#### 1. Networking Administrator

Role: Kumar is responsible for maintaining and managing the hospital's network infrastructure. They handle tasks such as configuring routers, switches, and other networking equipment, monitoring network performance, and troubleshooting network-related issues.

Responsibilities:

- Ensure high availability and reliability of network services.
- Conduct routine maintenance and updates of network equipment.
- Monitor network traffic to detect and resolve performance bottlenecks.
- Provide network access to users while adhering to access control policies.

Contribution: Choosing and integrating the right hardware and softwares to create the chosen network architecture and the specifications.

Impact: Kumar makes sure that operations are running smoothly. This is vital in a hospital setting, where network downtime can delay access to electronic medical records, disrupt communication between departments, and even impact life-saving equipment.

Value to Hospital Operations: Kumar contributes uninterrupted patient care, efficient workflow and reduces the delays in the treatment.

#### 2. Network Security

Role: Jie focuses on protecting the hospital's data and network resources from unauthorized access and cyber threats. He implements security protocols and regularly updates defenses against vulnerabilities.

Responsibilities:

- Monitor and respond to security incidents.

- Manage firewalls, intrusion detection systems (IDS), and intrusion prevention systems (IPS).
- Develop and enforce network security policies and procedures.
- Conduct security audits and risk assessments to maintain a secure environment for patient and hospital data.

Contributions: Guarding Patient Data and Privacy

Impact: Jie takes care of all personal information, to avoid the healthcare organization becoming one of the targets for the cyber threat landscape.

Therefore, the workload of the network security is critical with respect to data protection and regulatory compliance as well as the hospital's image in the healthcare industry and the society.

Value to Hospital Operations: By securing patient information, Jie maintains the privacy trust of the patients in the hospital. Our advanced cyber protection will assist the hospital's efforts to avoid such costly data breaches while maintaining patient privacy and confidentiality.

### **3. Network Architect**

Role: Chi-uba designed and planned the hospital's network infrastructure. He focused on creating a cost effective, scalable and flexible architecture to support current and future needs of the hospital which includes extending the network to other branches at different school campuses.

Responsibilities:

- Ensure network design supports high availability, redundancy, and scalability.
- Collaborate with other teams to integrate new technologies to the existing architecture.
- Plan and implement network infrastructure that aligns with the hospital's strategic goals.
- Evaluate and select appropriate networking solutions to optimize performance.

Contributions: Building the foundation for future growth of the hospital network.

Impact: As technology becomes more and more essential, hospitals count on IT to incorporate cutting-edge medical devices, link patient surveillance systems, and keep up with new healthcare programs. The network architecture staff Chi-uba ensures that the hospital's IT infrastructure can support these advances today and in the future.

Value to Hospital Operations: Chi-uba also keeps the hospital on the cutting edge of healthcare technology in the provision of modern services and planning with regard to future needs and instituting scalable solutions. It builds a solid groundwork that empowers the hospital to integrate newer tools and technologies that improve patient care and operational efficiency.

### **4. IT Support Specialist**

Role: Richard provides essential technical support to the hospital's staff and departments. He is the first point of contact for resolving hardware and software issues, ensuring that healthcare professionals can focus on patient care without interruption. He also generates proper documentation of the network created for proper maintenance and integration of new employees to the IT department.

Responsibilities:

- Troubleshoot and resolve IT issues for end-users, including hardware and software problems.
- Install, configure, and maintain computer systems and applications used by hospital staff.
- Manage and support user accounts, access permissions, and device setups.
- Generates documentation of the network procedures, architecture, maintenance routine, etc.
- Offer training and guidance to staff on best practices and new technology tools.

Contributions: Provides technical support to other non IT staff in the hospital and also aids in integration of new employees to the IT department.

Impact: Healthcare staff Richard rely on technology every day from accessing electronic records to operating diagnostic equipment. The IT support staff Richard resolves the technical issues and allows him to focus on patient care.

Value to Hospital Operations: Richard keeps downtime to a minimum in fixing critical equipment and ensures patient care with nominal interruptions, providing the flexible solutions that everyone calls on. Empowering medical and administrative staff to work well, knowing IT support is just a phone call away.

# CHAPTER 2

## Project Overview

This project simulates the design and implementation of a robust and secure network infrastructure for a hypothetical, but realistic, modern hospital. Our team, functioning as the hospital's IT department, will design, simulate, and test the network using Packet Tracer. This hands-on approach allows us to directly address the unique challenges of the healthcare sector, applying the knowledge we've gained throughout the course.

### 2.1 Defining Networking:

At its core, networking interconnects computing devices, enabling communication and resource sharing. Connecting computing devices over a large number gets more complicated and difficult when there is no communication structure or architecture. This led to the focus on network architecture in creating large networks. There are various network like software defined networking, star network, mesh network, bus network, ring network, etc. Effective networking requires careful consideration of hardware, software, protocols, security, and scalability – aspects we'll directly address in this project.

### 2.2 Justification for Choosing a Hospital:

The hospital setting presents a complex networking challenge. The high stakes of patient care, stringent regulatory requirements, and the need for seamless system integration offer an ideal scenario to showcase our understanding of network design principles. This project allows us to apply our knowledge of the network model's various layers in a practical context.

### 2.3 Networking Needs in a Hospital:

Hospitals demand high-performance, reliable, and secure networks. Their core functions—patient care, diagnostics, and administration—depend heavily on seamless connectivity. Specific needs include:

**High Availability:** Downtime has severe consequences. The network must ensure continuous access to Electronic Health Records (EHR), Picture Archiving and Communication Systems (PACS), and other mission-critical applications. Kumar, the network administrator, will focus on ensuring redundancy and failover mechanisms are in place.

**Data Security and HIPAA Compliance:** Hospitals handle sensitive patient data, requiring stringent security measures compliant with HIPAA. Jie, as our security specialist, will be

primarily responsible for implementing strong encryption, access controls, and regular security audits.

**Scalability:** Hospitals constantly evolve, requiring a network adaptable to future growth and technological advancements. Chi-uba, as the network architect, will ensure the design accounts for this by considering potential future needs in terms of device capacity and data traffic.

**Real-Time Data Transfer:** Applications like PACS demand rapid transfer of large medical images. Kumar, as the network administrator, will ensure that the network has the required bandwidth and low latency to support these real-time data transfer needs.

**Integration of Diverse Systems:** Hospitals utilize a multitude of medical devices and software. The network must integrate these systems seamlessly, enabling effective data sharing. This will be a crucial consideration for Chiuba in designing the overall network architecture.

## **2.4 Project Goals:**

Our primary project goals are: To design a secure, highly available network architecture for a hospital, fully compliant with HIPAA. (Chiuba, primary; Kumar, Jie contribute) To simulate this network in Packet Tracer, demonstrating functionality and performance. (Chiuba, primary; Kumar contributes) To test the network's performance using real-time and simulated testing, evaluating key metrics. (Chiuba, primary; Kumar, Richy contribute) To comprehensively document the entire process, from design to analysis. (All team members contribute)



# CHAPTER 3

## Network Architecture

This chapter details the network architecture designed by Chiuba for our chosen hospital. We have selected a three-tier client-server architecture, deemed optimal by Kumar and Jie after considering efficiency, scalability, and security requirements, all crucial factors in a hospital environment. This architecture is particularly suitable for managing the complex demands of robust data management, high availability, and stringent HIPAA-compliant security measures.

### 3.1 Architectural Design Rationale (Chiuba):

A three-tier architecture was chosen by Chiuba over simpler or more complex alternatives because it effectively balances critical factors for the hospital setting. The Core, Distribution, and Access layers of this architecture each have specific roles that are suited to a healthcare setting.

**Core Layer:** This layer serves as the backbone of the network, facilitating dependable, fast data transfer between departments. For continuous access to vital applications like Electronic Health Records (EHR) and medical imaging, redundant, high-capacity Cisco routers and switches guarantee little downtime and quick data transfer between departments.

**Distribution Layer:** This layer controls data flow and upholds security regulations. It is positioned between the core and access layers. This layer employs Layer 3 switches and VLANs to separate various hospital departments and applies Access Control Lists (ACLs) to ensure adherence to data privacy regulations. Through the reduction of superfluous traffic to the core, this segmentation also enhances performance.

**Access Layer:** This layer sits at the edge of the network and links endpoints like staff PCs, mobile devices, and patient monitoring systems. Layer 2 switches and wireless access points provide secure connectivity, and port security ensures only authorized devices access the network.

### 3.2 Existing Architectures Used By Other Hospitals

#### Single-MAN Architecture in Hospitals

Use Case: Smaller hospitals or clinics with basic networking needs.

Advantages:

**Cost Efficiency:** Smaller healthcare facilities may have budget constraints. A Single-MAN architecture allows them to set up a cost-effective network without the need for extensive infrastructure and maintenance associated with multiple networks.

**Simplicity:** For hospitals that do not require complex networking capabilities, a Single-MAN architecture provides straightforward management. IT staff can focus on maintaining one network without the challenges of coordinating multiple systems.

**Ease of Setup:** Implementing a Single-MAN is generally quicker and simpler, which is beneficial for smaller facilities that need to deploy their network swiftly to support operations and patient care.

Limitations:

**Reliability Risks:** A single point of failure can lead to significant downtime, which is critical in healthcare environments where immediate access to patient data is essential.

**Scalability Issues:** As the facility grows or expands services, they may face challenges in scaling the network without significant upgrades or investments.

### Dual-MAN Architecture in Hospitals

Use Case: Larger hospitals or healthcare networks that demand high availability and performance.

Advantages:

**High Availability:** Dual-MAN architectures provide redundancy, which is crucial for hospitals that cannot afford downtime. If one network fails, the other can seamlessly take over, ensuring that critical applications and patient data remain accessible.

**Enhanced Reliability:** Hospitals often rely on real-time data for patient care. A Dual-MAN setup minimizes the risk of service interruptions, which is vital for operations like surgeries, emergency services, and patient monitoring.

- **Load Balancing:** With two networks, hospitals can distribute data traffic more effectively, enhancing performance, especially during peak times. This is particularly important for large facilities that manage a high volume of patient data and real-time imaging.

- **Scalability:** As hospitals grow and adopt more technology (e.g., telemedicine, advanced imaging, and patient management systems), a Dual-MAN architecture can accommodate increased demand without requiring a complete overhaul of the existing network.
- **Compliance and Security:** With regulations like HIPAA in the U.S. mandating the protection of patient data, a more robust network architecture allows for better implementation of security measures, including data encryption and access controls, to safeguard sensitive information.

Limitations:

**Higher Costs:** The initial investment and ongoing maintenance for a Dual-MAN can be significantly higher, which may not be feasible for smaller hospitals.

**Complex Management:** Managing multiple networks can require more advanced IT resources and training, potentially straining the hospital's IT capabilities.

### Cloud-Based Network Architecture

- **Description:** In a cloud-based architecture, some hospitals utilize cloud providers to host applications, data, and sometimes even part of the network infrastructure. This approach offloads some processing and storage to cloud servers, reducing on-site infrastructure.
- **Use in Hospitals:** Some hospitals may adopt cloud-based architecture for patient data storage, collaboration tools, or telehealth applications. This model is especially useful for hospitals with limited IT resources or those looking to support remote access.
- **Reason for Choice:** Cloud-based architecture offers cost savings on infrastructure, scalability, and off-site backups for disaster recovery. It also supports remote access for healthcare professionals, an advantage for hospitals with satellite locations or telemedicine services.

### **3.2 Network Topology and Components (Kumar):**

Kumar designed the hospital network using a star topology at the physical layer, connecting all devices to central switches and routers. This topology simplifies management, enhances scalability, and facilitates fault isolation. The core network utilizes high-bandwidth, high-availability switches and routers capable of handling substantial data traffic. Redundancy at the core layer minimizes the impact of potential failures. **Physical Layer (Kumar):** A combination of fiber optic cabling (for

high-bandwidth connections) and Category 6 (or higher) twisted-pair cabling (for shorter distances) was specified by Kumar. Wireless access points, configured by Jie, will provide secure wireless connectivity for mobile devices using robust security protocols (WPA2-Enterprise with 802.1X authentication). Data Link Layer (Kumar): Kumar selected IEEE 802.3 (Ethernet) for wired connections and IEEE 802.11ac (or newer) for wireless access. VLANs (Virtual LANs), configured by Kumar, will segment the network, enhancing security and isolation for different departments and user groups. Network Layer (Kumar): Kumar implemented IPv4 addressing with proper subnetting for efficient network management. Dynamic Host Configuration Protocol (DHCP) automates IP address assignment. Transport Layer (Kumar): Kumar specified both TCP (for reliable data delivery, like the EHR system) and UDP (for low-latency applications prioritizing speed over reliability, such as real-time video conferencing). Application Layer (Kumar): Hospital applications will be deployed across the application servers, ensuring seamless access and integration. Kumar will ensure the applications utilize appropriate security protocols (e.g., HTTPS) to protect sensitive data.

### **3.3 Security Considerations (Jie):**

Jie implemented robust security measures given the sensitivity of patient data: Firewalls (Jie): Multiple layers of firewalls protect the network from unauthorized access. Intrusion Detection/Prevention Systems (IDS/IPS) (Jie): These systems monitor network traffic for malicious activity and mitigate threats. Access Control Lists (ACLs) (Jie): ACLs, configured by Jie on routers and switches, restrict network access based on user roles and departments. Virtual Private Networks (VPNs) (Jie): VPNs secure remote access for authorized users. HIPAA Compliance (Jie): All aspects adhere to HIPAA regulations for patient privacy and data security. Regular security audits and penetration testing will be conducted by Jie to ensure ongoing network security.

# Chapter 4

## Network Implementation

The chapter discusses the practical implementation of the new network infrastructure of the hospital, elaborating on the design plan that was developed in Chapter 5. In this chapter, the implementation plan is generally centered on a phased rollout to minimize disruption of activities at the hospital. In the architecture of the network, a client-server model is followed with three tiers.

**Phase 1: Core Network Infrastructure** In this stage, the main emphasis is on setting up the components of the network. Getting the hardware and setting it up involves acquiring and installing the network components, such as; We plan to use Cisco 2811. Cisco 2911 routers to facilitate routing between network segments, like the internal network and the Internet within the hospital network setup. We aim to set up routing protocols such as OSPF and inter-VLAN for dependable routing operations in the hospital network system. The Cisco MULTILAYER 3650 switches will be used to interconnect network devices in each department to ensure reliable data transmission at high speeds using technologies such as 10 Gigabit Ethernet and PoEs for IP phones, among others. VLAN implementation is planned for enhancing security measures and segregating network traffic purposes. Firewall security measures are put in place using SSH to safeguard the network's boundaries and prevent entry. Policies will be set up to permit access while barring any traffic. Setting up cabling and connections includes the installation of the cabling network infrastructure, such as utilizing fiber optic cables, which is essential for establishing high-speed connections between systems, like PAC servers and the central network core, meeting the demands of imaging effectively.

**Phase 2: Departmental Network Implementation** The CISCO 2960 switches in the department will be linked to the switches through a combination of fiber optic cables and UTP Category 6A cables as needed. Incorporating equipment into the system involves connecting patient monitors and diagnostic tools while prioritizing network security and following manufacturer guidelines and quality of service standards diligently. For enhanced network segregation between departments, each department was assigned a single VLAN network. The hospital has 9 various departments as follows:

- Administration and Support Department (ASD)
- Medical and Surgical Department (MSD)
- Diagnostic and Imaging Department (DID)
- Therapeutic and Rehabilitation Department (TRD)

- Special Care Department (SCD)
- Patient and Family Support Department (PFSD)
- Pharmacy Department (PD)
- Infection Control And Epidemiology (ICE)
- Visitors and Guests (VG)

Phase 3: Wireless Network Deployment This phase involves the implementation of a secure wireless network using access points: Access Point (AP) Selection: The hospital will utilize Cisco Aironet 1815i to provide reliable and secure wireless connectivity.

#### **4.1 CISCO Code Implementation (Packet Tracer)**

##### **Layer 2 Switch Config:**

```

en
conf t
hostname ASD-SW
enable password hospital
banner motd #Unauthorized Access#
no ip domain lookup
line console 0
password hsp
login
ex
service password-encryption

```

##### **Layer 3 Switch Config:**

```
conf t
hostname MultiSW1
enable password hospital
banner motd #Unauthorized Access#
no ip domain lookup

line console 0
password hsp
login
exit

service password-encryption
```

```
ip domain name hospital.net
username admin password hsp123
crypto key generate rsa
1024

line vty 0 15
login local
transport input ssh
exit
```

## **VLAN CONFIG**

```
vlan 20
name MSD
```

exit

int range fa0/1-2

switchport mode trunk

exit

int range fa0/3-24

switchport mode access

switchport access vlan 20 exit

(note: Similarly other VLAN upto VLAN 90 for Visitor and Guest)

## **Multilayer Switchport VLAN**

vlan 10

vlan 20

vlan 30

vlan 40

vlan 50

vlan 60

vlan 70

vlan 80

vlan 90

exit

int range gig1/0/2-10

switchport mode trunk

exit



do wr

## **OSPF Routing**

ip routing

router ospf 10

network 192.168.100.0 0.0.0.63 area 0

network 192.168.100.64 0.0.0.63 area 0

network 192.168.100.128 0.0.0.63 area 0

network 192.168.100.192 0.0.0.63 area 0

network 192.168.101.0 0.0.0.63 area 0

network 192.168.101.64 0.0.0.63 area 0

network 192.168.101.128 0.0.0.63 area 0

network 192.168.101.160 0.0.0.63 area 0

network 192.168.101.192 0.0.0.63 area 0

network 192.168.102.80 0.0.0.3 area 0

exit

ip route 0.0.0.0 0.0.0.0 192.168.102.86 #backup ISP

ip route 0.0.0.0 0.0.0.0 192.168.102.81 70 #main ISP

# CHAPTER 5

## Project Design

### 5.1 Planning

We wanted the network to fit a few requirements that would help the hospital's needs. some of the requirements we thought about are as follows:

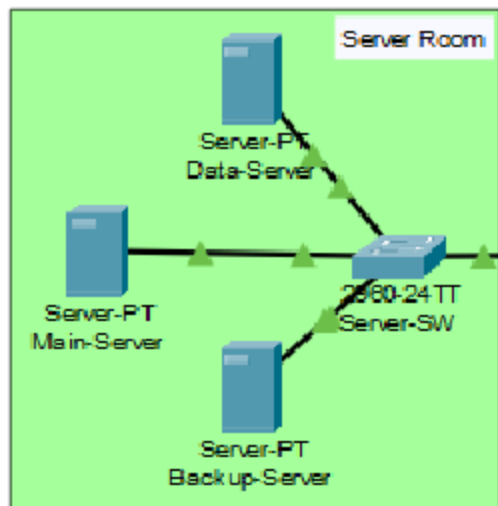
- Cost-effective: the network installation and management should be cost-effective, providing good results at substantially lower prices than the market.
- Scalable and allows for future branch expansion and implementation.
- Consistent and reliable network with little to no downtime.
- Fast and high bandwidth for important medical devices.
- Network for guests and visitors of patients.
- Specialized access and security measures to prevent unauthorized access to sensitive data.
- WiFi enabled.

### 5.2 Proposed Network

The network requirement that we found suitable to fit the above mentioned constraints are as follows:

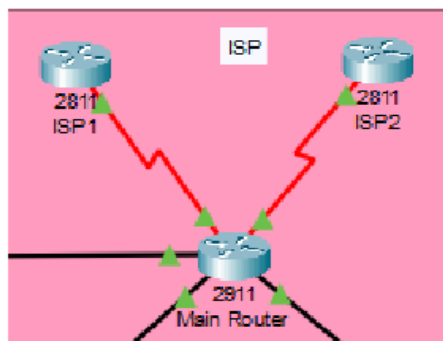
- A three tier model to enhance scalability in addition of various departments or possibly even a new branch from a different remote location.
- Two ISP (Internet Service Providers), to have almost no network down-time to ensure fast and reliable data transfers for critical information.
- 10-bit Gigabit Ethernet cable installation to allow high speed network access and flow.
- Using VLANs to divide respective departments but at the same time having inter-VLAN routing to allow communication between various said departments.
- Configuring load balancing and traffic ip address hopping to prevent choke points or latency in network.

### 5.3 Building Network

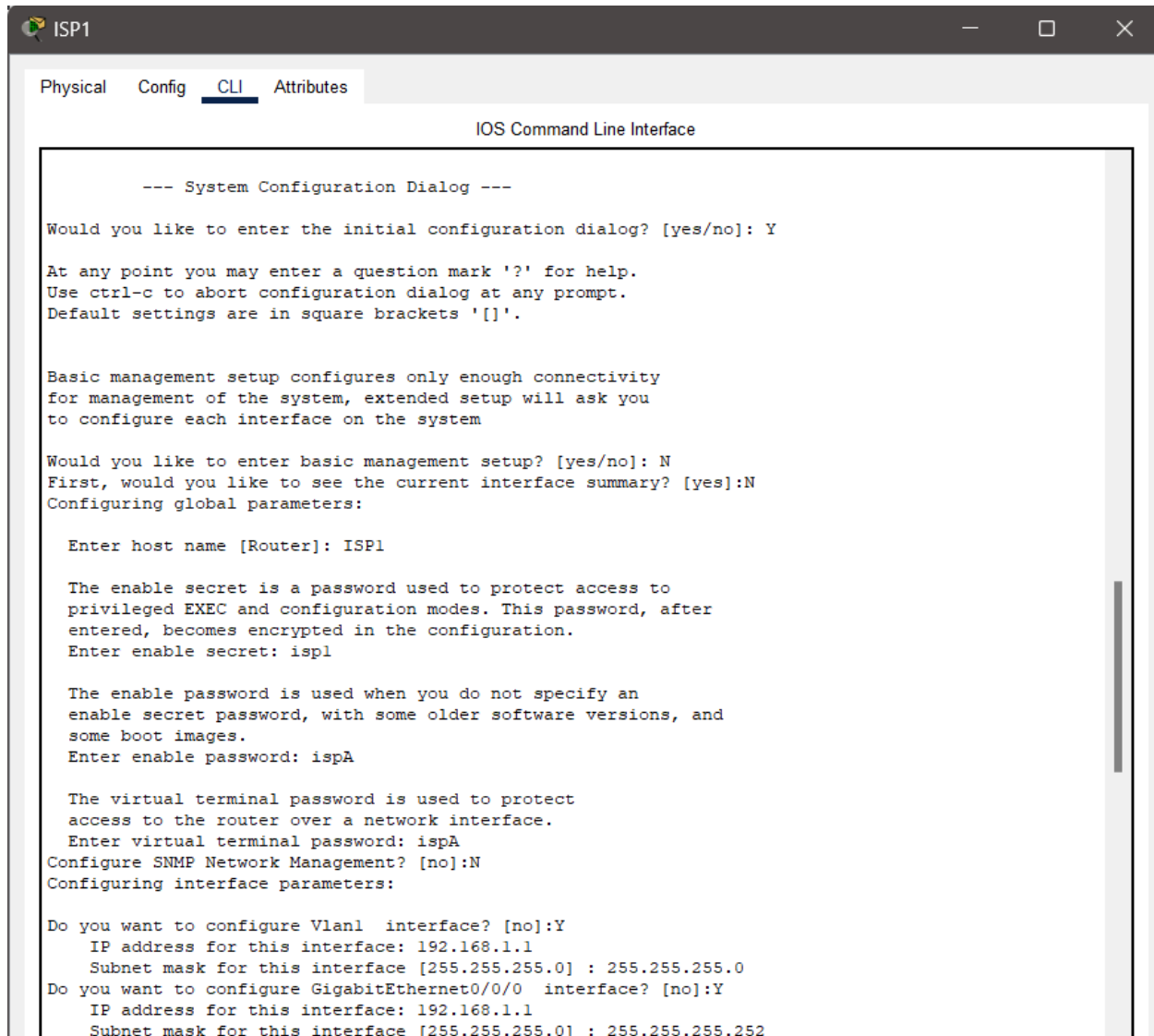


(fig 5.1)

Creation of a server room having three servers for data, main communication and a backup relay server is shown in *fig 5.1*.



Creation of 2 ISP providers to ensure little to no network downtime. One acts as the primary provider while the other is used as a backup to ensure fast switching of ISP if primary network is down.

The image shows a window titled 'ISP1' with a dark header bar. Below the header, there are four tabs: 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is selected and highlighted. The main content area is titled 'IOS Command Line Interface' and displays a text-based configuration dialog. The dialog starts with '--- System Configuration Dialog ---' and asks if the user wants to enter the initial configuration dialog, with 'Y' entered. It then provides instructions on using '?' for help, ctrl-c to abort, and square brackets for defaults. Next, it asks if the user wants to enter basic management setup, with 'N' entered. It then asks if the user wants to see the current interface summary, with 'N' entered. The configuration proceeds to global parameters, where the host name is set to 'ISP1'. It then asks for an enable secret password, with 'ispl' entered, and an enable password, with 'ispA' entered. It then asks for a virtual terminal password, with 'ispA' entered. It then asks if the user wants to configure SNMP Network Management, with 'N' entered. Finally, it asks if the user wants to configure interface parameters. For the 'Vlan1' interface, it asks for an IP address (192.168.1.1) and a subnet mask (255.255.255.0). For the 'GigabitEthernet0/0/0' interface, it asks for an IP address (192.168.1.1) and a subnet mask (255.255.255.252).

```
ISP1

Physical  Config  CLI  Attributes

IOS Command Line Interface

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: Y

At any point you may enter a question mark '?' for help.
Use ctrl-c to abort configuration dialog at any prompt.
Default settings are in square brackets '[]'.

Basic management setup configures only enough connectivity
for management of the system, extended setup will ask you
to configure each interface on the system

Would you like to enter basic management setup? [yes/no]: N
First, would you like to see the current interface summary? [yes]:N
Configuring global parameters:

  Enter host name [Router]: ISP1

  The enable secret is a password used to protect access to
  privileged EXEC and configuration modes. This password, after
  entered, becomes encrypted in the configuration.
  Enter enable secret: ispl

  The enable password is used when you do not specify an
  enable secret password, with some older software versions, and
  some boot images.
  Enter enable password: ispA

  The virtual terminal password is used to protect
  access to the router over a network interface.
  Enter virtual terminal password: ispA
Configure SNMP Network Management? [no]:N
Configuring interface parameters:

Do you want to configure Vlan1 interface? [no]:Y
  IP address for this interface: 192.168.1.1
  Subnet mask for this interface [255.255.255.0] : 255.255.255.0
Do you want to configure GigabitEthernet0/0/0 interface? [no]:Y
  IP address for this interface: 192.168.1.1
  Subnet mask for this interface [255.255.255.0] : 255.255.255.252
```

Configuration of IP and Subnet masking for the above 2 ISPs. We designed the ISP to provide a good connection while also incorporating a large number of device capability per VLAN. The way of determining subnet masking was by applying the formula  $2^{\text{number of host bits} - 2}$ . We chose a subnet mask of 255.255.255.252 for the hosts as it allows the allocation of two IP addresses for the ISPs ensuring only 2 hosts (primary and backup).

Rest of the end devices and switches have been assigned a subnet mask of 255.255.255.128 to provide upto 62 allocatable IP addresses ensuring enough devices per department.

Main-Server

Physical
Config
**Services**
Desktop
Programming
Attributes

**SERVICES**

HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP

DHCP

Interface
FastEthernet0
Service
☒ On
☐ Off

Pool Name
serverPool

Default Gateway
0.0.0.0

DNS Server
0.0.0.0

Start IP Address :
192
168
102
0

Subnet Mask:
255
255
255
0

Maximum Number of Users :
256

TFTP Server:
0.0.0.0

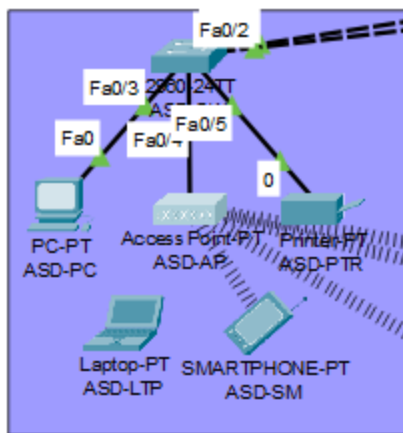
WLC Address:
0.0.0.0

Add
Save
Remove

| Pool Name  | Default Gateway | DNS Server | Start IP Address | Subnet Mask  | Max User | TFTP Server | WLC Address |
|------------|-----------------|------------|------------------|--------------|----------|-------------|-------------|
| serverPool | 0.0.0.0         | 0.0.0.0    | 192.168.1...     | 255.255.2... | 256      | 0.0.0.0     | 0.0.0.0     |
| VGPpool    | 192.168.1...    | 0.0.0.0    | 192.168.1...     | 255.255.2... | 22       | 0.0.0.0     | 0.0.0.0     |
| ICEPool    | 192.168.1...    | 0.0.0.0    | 192.168.1...     | 255.255.2... | 22       | 0.0.0.0     | 0.0.0.0     |
| PDPool     | 192.168.1...    | 0.0.0.0    | 192.168.1...     | 255.255.2... | 57       | 0.0.0.0     | 0.0.0.0     |
| PFSDPool   | 192.168.1...    | 0.0.0.0    | 192.168.1...     | 255.255.2... | 57       | 0.0.0.0     | 0.0.0.0     |
| SCDPool    | 192.168.1...    | 0.0.0.0    | 192.168.1...     | 255.255.2... | 57       | 0.0.0.0     | 0.0.0.0     |
| TRDPool    | 192.168.1...    | 0.0.0.0    | 192.168.1...     | 255.255.2... | 57       | 0.0.0.0     | 0.0.0.0     |
| DIDPool    | 192.168.1...    | 0.0.0.0    | 192.168.1...     | 255.255.2... | 57       | 0.0.0.0     | 0.0.0.0     |
| MSDPool    | 192.168.1...    | 0.0.0.0    | 192.168.1...     | 255.255.2... | 58       | 0.0.0.0     | 0.0.0.0     |

☐ Top

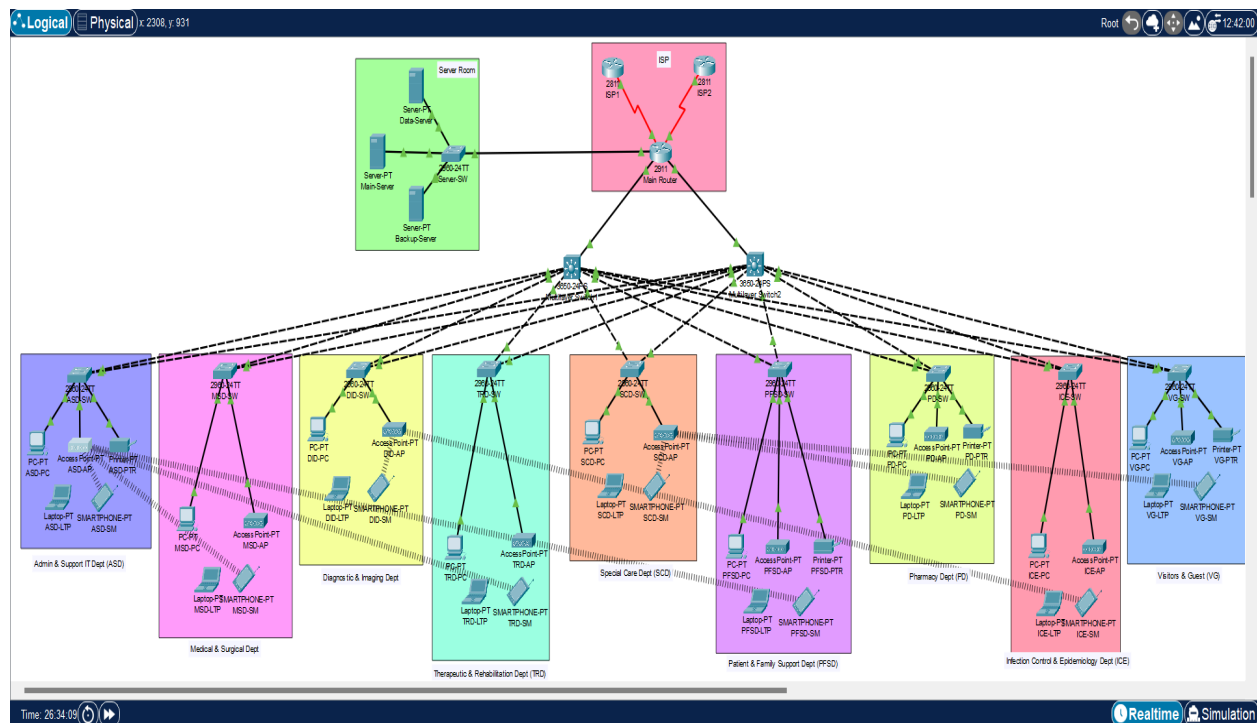
Creation of a network pool to dynamically assign DHCP IP address for various VLANs. When a device connects to it, the server lends an IP from this pool, alongside other network configuration details, such that seamless and automated network access is allowed without necessarily having to execute any form of IP configuration manually.



Admin & Support IT Dept (ASD)

Creation of a single department with a switch, access point, and various end devices.

*Note: this process is replicated for the other eight departments*

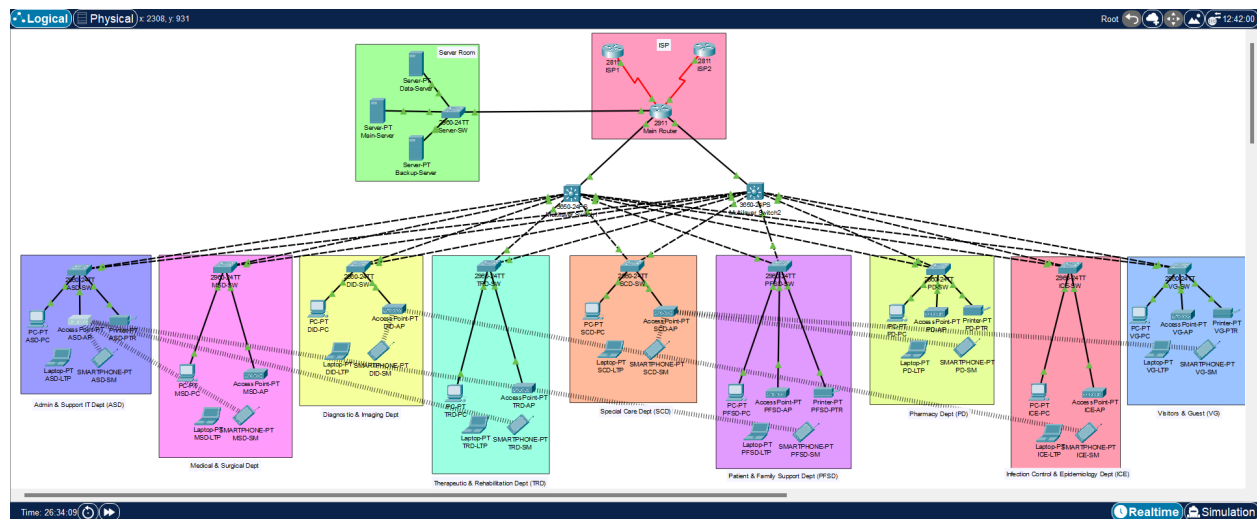


Final network architecture.

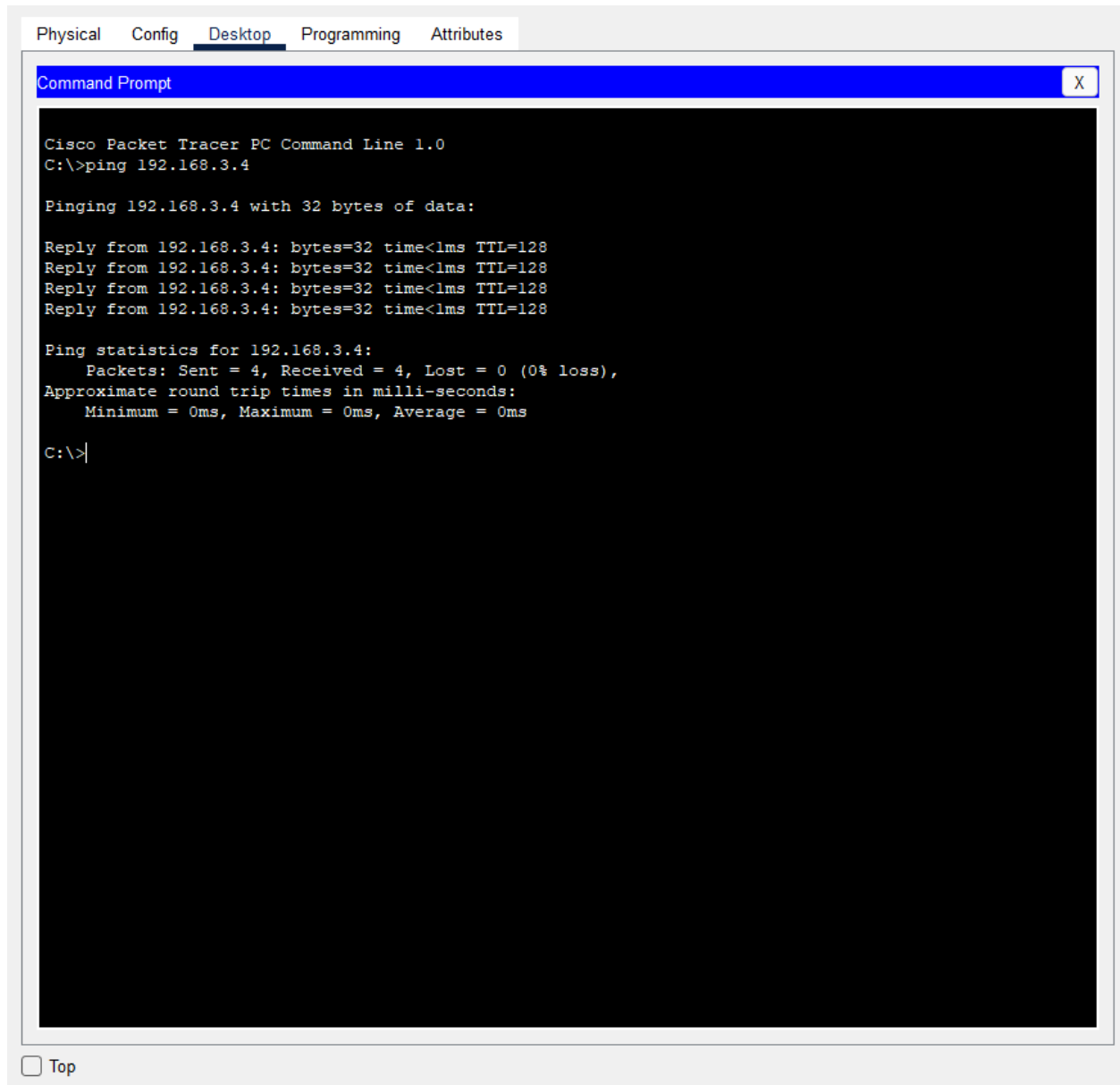
# Chapter 6

## Network Testing

This chapter details the testing procedures employed to evaluate the performance and functionality of the designed hospital network, both through real-time testing (where applicable) and simulation using Cisco Packet Tracer. The goal is to assess the network's stability, speed, security, and overall effectiveness in meeting the hospital's requirements. (network architecture) (dhcp pool) (real time - ping test) (simulation - packet transmission) (inter-department communication)

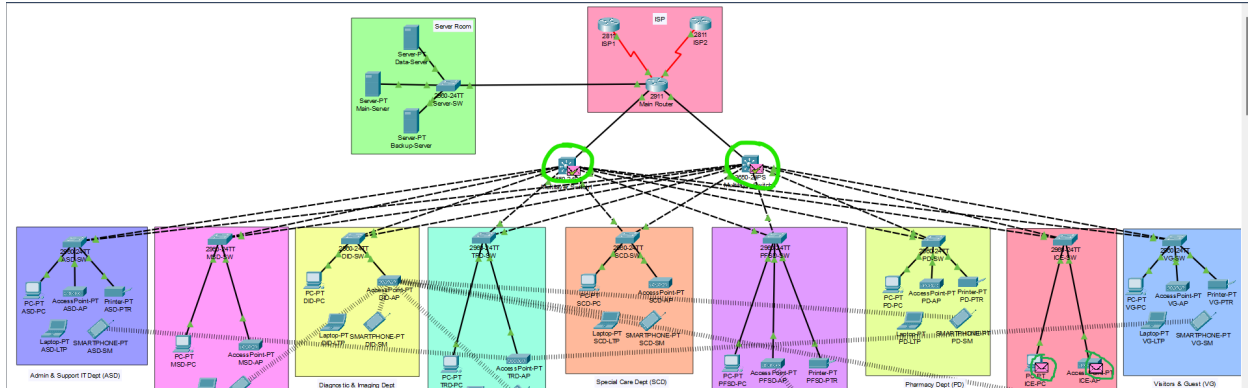


The above diagram shows how the final network architecture is. We shall now run a simple ping test between two departments to show the working (between Infection Control and Epidemiology department & Therapeutic and Rehabilitation department).

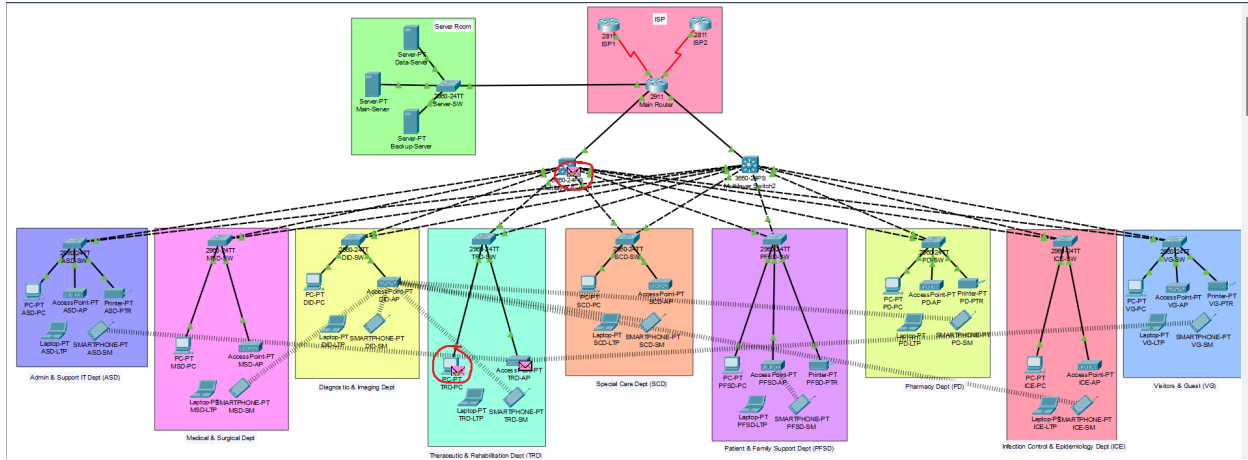


We go into the command prompt (cmd) of a given PC of the Infection Control and Epidemiology department and run a simple ping test to check transmission of packets.





We notice that the packets are going from the PC of Infection Control and Epidemiology department to that of the multilayer third level switches which will further handle the routing of data packets to the required destination (Therapeutic and Rehabilitation department in this example).



The data packets are received to the end devices of the Therapeutic and Rehabilitation department. This ensures reliable transmission and good connectivity.

Similarly, we can use this network architecture to ping various devices either between departments or internally ensuring reliable communication necessary for hospitals.

# Chapter 7

## Conclusion

This project has provided valuable insights into the complexities and considerations of designing a network architecture, particularly for a hospital setting with its unique demands and constraints. The hands-on experience of implementing the network in Cisco Packet Tracer has deepened our understanding of network components, configurations, and troubleshooting.

The project highlighted the critical importance of network security in a healthcare setting, where sensitive patient data must be protected from unauthorized access and cyber threats. Implementing multiple layers of security, including firewalls, intrusion detection systems, and access controls, underscored the need for a comprehensive approach to network security.

Furthermore, the project emphasized the importance of collaboration and teamwork in network design and implementation. Each team member's expertise contributed to the successful completion of the project, demonstrating the value of diverse skillsets in addressing complex networking challenges.

Overall, this project has been a rewarding learning experience, providing practical insights into network design principles and their application in a real-world scenario. The knowledge and skills gained will be invaluable in our future careers as networking professionals.

## 7.1 Future Work for the Hospital IT Department

### 1. Advancing Telehealth and Patient Monitoring:

We are innovative thinkers in the creation of systems to advance telemedicine toward closing the gaps in direct patient care. Thus, we are able to enhance our network to accommodate video teleconferencing, easily accessible secure connections, and integrate remote monitoring equipment capable of transmitting

secure data.

## 2. Advancing Patient Care with Data Insights and AI:

More information is tending to seize control of the determination of patient needs and forecasting health outcomes. We have some potentials to introduce advanced data analytics to contend with AI tools, which will help our clinicians make approach decisions and personalize care.

## 3. Adapting Cybersecurity to Evolving Risks:

An essential opportunity to make patient data secure, particularly as schemes of any particular cyber threat sharpen. The ongoing development of our security strategies to include stronger network access controls, AI-based threat detection, and periodic training is underway, systemic and culture-wide. The steps help us ensure preparedness and vigilance, illuminating the ever-changing landscape of cybersecurity.

## 4. Boosting EHR Connectivity Across Systems

Seamless data exchange between healthcare providers is critical to comprehensive patient care. Coordination toward achieving compatible systems among healthcare providers through standardized data formats and secure application programming interfaces is being done to share essential patient information fast and securely.

## 5. Adopting Cloud Technologies for Enhanced Flexibility and Savings

As 5G networks roll out, we're looking at leveraging faster speeds and better connectivity for telemedicine and real-time imaging. We're planning to collaborate with providers to integrate 5G, ensuring it meets our needs and security standards.

## 6. Accelerating IT Assistance for Greater Accessibility:

To serve our staff better, we're exploring options for self-service IT solutions. Simple things like resetting passwords or basic troubleshooting could be automated or handled through an easy-to-use portal, allowing our team to

respond faster to more complex issues.

7. Integrating Advanced Technologies in Healthcare:

Smart hospitals" utilize digital wayfinding, asset tracking, and room automation. We're designing a secure, efficient network for these technologies to enhance patient experience and streamline operations.

8. Ready for 5G: Unlocking New Possibilities in Healthcare:

As 5G networks roll out, we're exploring how faster data speeds and better connectivity could help in areas like telemedicine and real-time imaging. We're working with providers to integrate 5G in a way that aligns with our needs and security standards.

9. Improving Wireless Connectivity Efficiency:

We're boosting our wireless network to make it easier for hospital staff to stay mobile without compromising security. This might include using cutting-edge tech like Wi-Fi 6E and mesh networks.

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